

Mini-Projet IA for Software Engineering

Subject ideas for the team - A.U 2026-2027 - Draft for discussion

What the teacher asks for

- A **web application** (front-end + back-end) with an **AI module**. Groups of 5-6.
- Week 1: subject approved. Week 3: progress check. Week 7: final code + report + 15-min live demo.
- Each member's grade can be adjusted by their **Git commits**, so everyone needs their own part.
- LLM rules: no raw personal data sent to the LLM, check the AI output before showing it, handle API errors (timeout, quota, missing key).
- Deliverables: Git repo, working demo, presentation of the prompts used, technical report.

How we will work (same for every idea)

- **Stack:** Spring Boot (back) + Angular (front) + MySQL. One single app split into 5 modules (no microservices).
- **Each member owns 1 business module:** its Angular pages, its Spring endpoints and its own database tables.
- **Shared AI module:** one common *AiService* (free LLM: Groq, Gemini, Mistral or local Ollama). Each member adds **1 AI feature** to their module and **1 skill** to a shared chat assistant.
- **Nobody waits for anybody:** in week 1 we agree on the entities, the APIs and some sample data. Modules only link to each other by ID. The AI has a 'fake mode' so everyone can code before any API key works.

The 8 ideas

1. RecrutAI - Recruitment platform

Extends the CV example from the brief into a full hiring tool. HR tech is in demand in Tunisia and the EU.

#	Module (front + back)	AI feature
1	Job offers (create, publish, archive)	Generate a job description from 5 fields
2	Candidates & CVs (profile, PDF upload)	Extract skills + 3-sentence summary (personal data masked)
3	Applications (apply, status pipeline)	Candidate / offer match score out of 100, with an explanation
4	Interviews (scheduling, evaluation grid)	Interview questions tailored to the offer + candidate
5	Company & HR dashboard	Automatic weekly summary of recruiting activity

2. ClinicFlow - Private clinic management

Useful but medical data is sensitive: needs strong anonymization and disclaimers.

#	Module (front + back)	AI feature
1	Patients (record, history)	Summary of the medical history
2	Appointments (calendar, doctors)	Urgency level suggested from the symptoms
3	Consultations & prescriptions	Consultation report written from the doctor's notes
4	Pharmacy / stock	Low-stock alerts + reorder suggestion
5	Billing & dashboard	Monthly business summary

3. SouqAI - Marketplace for Tunisian artisans

Visual and appealing, but looks like many student projects.

#	Module (front + back)	AI feature
1	Products & categories	Product description + tags in FR/EN/AR
2	Shops & sellers	Shop 'about' page generated + profile advice
3	Cart & orders	'Frequently bought together' recommendations
4	Customer reviews	Sentiment analysis + summary of reviews per product
5	Delivery & dashboard	Summary of sales + reasons for delays

4. DeskAI - IT helpdesk and support tickets

Like GLPI / Jira Service Desk, used by most Tunisian companies and banks. Easiest for beginners.

#	Module (front + back)	AI feature
1	Tickets (create, status, priority)	Auto-classify ticket category + urgency from its text
2	Knowledge base (articles, FAQ)	Suggest articles that answer a new ticket
3	Agents & teams (assignment, workload)	Recommend the best agent for the ticket
4	IT equipment (PCs, licenses, loans)	Summary of each device's incident history
5	Deadlines & dashboard	Weekly report + detection of recurring problems

5. ImmoAI - Real estate agency

Visual and concrete; the natural-language search makes a great demo.

#	Module (front + back)	AI feature
1	Listings (properties, photos)	Listing text generated in FR/EN/AR
2	Clients & needs (buyers, renters)	Free text to search criteria ('S+2 in La Marsa under 1500 DT')
3	Visits (scheduling, visit report)	Clean visit report from the agent's rough notes
4	Contracts & payments	Summary of a contract's key clauses
5	Dashboard & price estimate	Price estimate from similar properties + explanation

6. TripAI - Travel agency / Tunisian tourism

Tourism is a key Tunisian sector; the generated itinerary gets attention in a demo.

#	Module (front + back)	AI feature
1	Trip packages (destinations, prices)	Descriptions generated in several languages
2	Bookings	Custom day-by-day itinerary from budget and preferences
3	Partners (hotels, transport, guides)	Pick partners suited to each trip
4	Customer reviews	Sentiment analysis + summary per trip
5	Customers & dashboard	Personalized trip recommendations + season summary

7. FleetAI - Transport and delivery fleet

Real need for Tunisian SMEs, but less visual.

#	Module (front + back)	AI feature
1	Vehicles (fleet, documents)	Alerts before insurance / inspection expires
2	Drivers (licenses, availability)	Driver assigned to a mission with a justification
3	Missions / deliveries	Delivery order written from a free-text request
4	Maintenance	Diagnosis suggested from the fault description
5	Fuel & costs dashboard	Detect abnormal fuel consumption + monthly report

8. FitAI - Gym / fitness club

Fun and easy for beginners, but weighs less on a CV.

#	Module (front + back)	AI feature
1	Members & subscriptions	Detect members likely to quit + message to keep them
2	Classes & coaches (schedule)	Suggest classes that fit a member's goals
3	Training programs	Personalized weekly training program
4	Nutrition	Meal plan from the member's goal and budget
5	Equipment & dashboard	Equipment maintenance + club activity summary

Comparison (1 = weak, 5 = strong)

Idea	CV value (Tunisia/EU)	Beginner-friendly	Demo	AI usefulness
1 RecrutAI (hiring)	5	4	4	5
4 DeskAI (IT support)	5	5	4	5
5 ImmoAI (real estate)	4	4	5	4
6 TripAI (tourism)	3	4	5	4
7 FleetAI (fleet)	4	3	3	3
8 FitAI (gym)	3	5	4	4
3 SouqAI (marketplace)	3	4	4	3
2 ClinicFlow (clinic)	4	3	3	4 (sensitive data)

Top 2 suggestions: RecrutAI (best story, based on the teacher's example) or DeskAI (easiest, AI works reliably even with free models).

Simple 7-week plan

Week	Goal
1	Pick the subject, get it approved. Together: agree on entities, APIs, sample data. Set up the skeleton + Git repo.
2	Each member: CRUD of their module (Spring endpoints + Angular pages) using sample data.
3	Progress check: all 5 modules work on their own; each AI feature works in fake mode.
4	Connect the real free LLM. Each member writes and tests their prompt.
5	Link modules together by ID. Add each member's skill to the shared chat assistant.
6	Tests, bug fixing, deployment, write the report (each member writes their module's section).
7	Rehearse the demo twice, freeze the code, final defense.

Next step

Each member picks their top 2 ideas. We choose one, then split the 5 modules between us.